

Mahriar Gharaghani

Artificial Intelligence | Operations Research | Decision Intelligence

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Scholar (optional): **TODO:** `scholar.google.com/...`

Research Interests

Sequential decision-making under uncertainty; reinforcement learning for operations; optimization-informed AI systems; project portfolio optimization and budgeting under cash-flow constraints; simulation–optimization.

Education

M.Sc. in Industrial Engineering **TODO:** 2023–2026

Iran University of Science and Technology (IUST), Tehran, Iran

Thesis: *Developing an AI Agent for Project Portfolio Budget Allocation under Uncertain and Dynamic Cash-Flow Constraints*

Supervisor: **TODO:** Seyed Farid Ghannadpour (if official)

TODO: GPA (optional), relevant coursework (optional)

B.Sc. in TODO: Industrial Engineering / ... **TODO:** Start–End

TODO: University name, City, Country

TODO: GPA (optional), relevant coursework (optional)

Research Experience

M.Sc. Thesis Researcher, Iran University of Science and Technology (IUST) **TODO:** 2023–2026

AI Agent for Project Portfolio Budget Allocation under Uncertain Cash-Flow Constraints

- Developed a sequential decision-making formulation for multi-period portfolio budgeting under uncertain and dynamic cash-flow constraints.
- Integrated optimization, simulation, and AI-based policy learning to generate feasible allocation decisions under uncertainty.
- Designed an evaluation protocol comparing learned policies to a perfect-foresight optimization benchmark as an upper bound.
- **TODO:** Add concrete methods: MIP/DP, scenario simulation, RL algorithm, environment design, reward, constraints handling.

Externally Supported Thesis Context **TODO:** Year(s)

Supported under a contract with **Tehran Industrial Estates Company**.

- Aligned problem formulation and evaluation with real institutional project portfolio and budgeting needs.
- **TODO:** Add what you can safely claim (data type, validation, deliverables).

Publications

TODO: If none, delete this section entirely.

- **TODO:** Author(s). Title. Venue, Year.

Selected Projects

TODO: Simulation–Optimization Prototype for Cash-Flow Feasibility **TODO: Year**

- **TODO:** What you built, methods used, and outcome.
- Tech: **TODO:** Python, Pyomo, Gurobi/CPLEX, NumPy/pandas, ...

TODO: RL Baselines for Sequential Budget Allocation **TODO: Year**

- Implemented baseline policies and an evaluation harness for sequential allocation under uncertainty.
- **TODO:** Algorithms: PPO/DQN/A2C; custom environment; metrics; ablations.

Professional Experience

Project Control / Planning Engineer, TODO: Company (Oil & Gas) **TODO: Start–End**

- Worked on project-oriented decision problems involving planning, cost control, and schedule/budget monitoring in industrial environments.
- **TODO:** Add 2–4 strong bullets with measurable outcomes and analytical/automation components if any.

Technical Skills

Programming: Python, **TODO: MATLAB**, **TODO: SQL**

Optimization / OR: **TODO:** Pyomo, **TODO:** Gurobi/CPLEX, **TODO:** OR-Tools, mixed-integer optimization, simulation

Machine Learning / AI: **TODO:** PyTorch, **TODO:** scikit-learn, reinforcement learning (**TODO:** Stable-Baselines3 / RLlib)

Tools: Git, Linux, L^AT_EX

Languages

Persian (Native); English (**TODO:** Professional working proficiency / Full professional proficiency)

References

Available upon request.

Seyed Farid Ghannadpour, Associate Professor, Department of Industrial Engineering, Iran University of Science and Technology

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